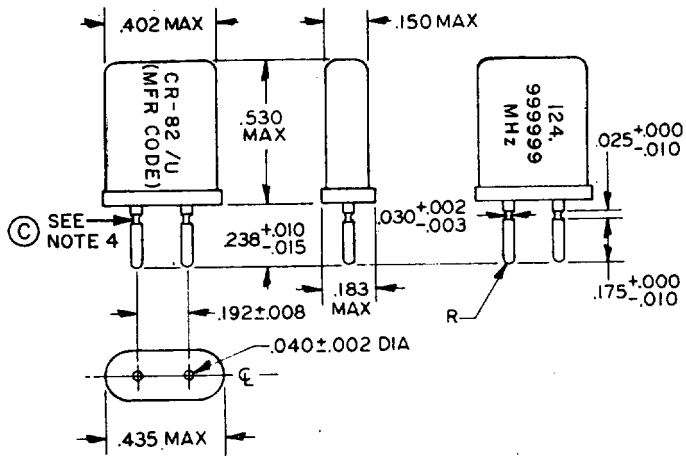
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-82/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth overtone; noncontrolled; series.



INCHES	MM	INCHES	MM
.002	.05	.150	3.81
.003	.08	.175	4.45
.008	.20	.183	4.65
.010	.25	.192	4.88
.015	.38	.238	6.05
.025	.64	.402	10.21
.030	.76	.435	11.05
.040	1.02	.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designation, frequency, and code designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
© 4. The pin undercut may be omitted.

© denotes change

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-25/U (MIL-H-10056/11)
Dimensions, marking, and configuration	See figure
Frequency range	50 to 125 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Frequency tolerance (operating temperature range)	±50 PPM
Equivalent resistance	MHz / Ohms, maximum 50 to 90 / 50 90+ to 125 / 60
Mode of oscillation	Fifth mechanical overtone
Government reference standard test sets	TS-683/TSM or AN/TSM-15
Calibration values:	
Resistance	60 ohms
Resistor voltage drop	0.35 volt
Drive level, rated	2.0 ± 0.4 mW
Operating temperature range	-55° to +105°C, inclusive
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration high frequency: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 204 of MIL-STD-202, test condition A.

NOTE: NATO preferred type NXT44.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0408-32)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-3098/59C CRYSTAL UNIT, QUARTZ, CR-82/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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